Ruiqi Ding

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Research Assistant, National University of Singapore (**NUS**)
Studied Physics to understand **Nature**, now exploring Neuroscience to understand **Intelligence**

1 Education

- 2016 - 2020, **NUS**, B.Sc. (1st Class Honours) Physics, GPA 4.52/5
- 2019, **University of Göttingen**, M.Sc. (Non-degree Exchange Program) Physics, GPA 1.4/1

2 Employment

- 2020 - now, **NUS** Gao lab, Research Assistant in Superconducting Quantum Computing

3 Research

- 2020 - now, **Stanford** Lin lab, Genetically Encoded Voltage Indicator Molecular Dynamics
  - Investigating kinetic barriers encountered by GEVI during activation/inactivation using MD simulation and analysis, in order to develop faster indicators of neural activity
- 2020 - now, **NUS** Gao lab, Quantum Optimal Control & RF Electronics & Various Lab Software
  - Developing Deep Reinforcement Learning based algorithm, wrote ODE solvers for QOC, in order to achieve fast and high fidelity quantum operation
  - Built RF electronics systems with ESP32 microcontroller, Raspberry Pi and custom PCBs. Wrote PID programs. Carried out calibrations like IQ mixer tuning, in order to drive/measure the superconducting Qubit
  - Developed Full Stack (MCU + Python & Node backend + React frontend App + WebRTC) real-time lab monitoring system, enabling lab members to monitor essential equipment like the Dilution Refrigerator anytime anywhere via the internet. Wrote programs & GUI to control Signal Generators / Spectrum Analyzers etc.
- 2019 - 2020, **NUS** Englert group, Density (Potential) Functional Theory
  - Developed DPFT program using the GPU parallel acceleration advantage of PyTorch. Used it to study Spin Polarized Fermi Gas with magnetic dipole-dipole interaction in both position & momentum space
- 2019, **Göttingen** Kree group, Deep Reinforcement Learning & Medical Nanorobots
  - Studied the physics of Complex Fluid (human tissue) and programmed it as a stochastic RL environment. Used RL to navigate the (simulated) nanorobot through complex fluid, in order to achieve medical applications such as Drug/Gene Delivery
- 2018 - 2019, **NUS** Matsukevich lab, Laser & FPGA Systems
  - Built laser system with PID control, for Laser Cooling and Trapping of Ytterbium Ion Qubit
  - Built FPGA system (Linux + Verilog + Websocket) for high speed signal processing, such as photon counting and quantum gate sequencing
- 2017 - 2018, **NUS** Englert group, Stern Gerlach Interferometer Quantum simulation
  - Wrote MATLAB & Mathematica programs to simulate & calculate the Stern Gerlach Interferometer, in order to benchmark a new 4th-order Suzuki-Trotter approximation to the unitary evolution operator, based on consecutive Fourier Transforms between position and momentum space
- 2016, **NUS** Sow lab, Carbon Nanotube growth
  - Learnt to grow Carbon Nanotubes. This was a learning oriented project with no contributions achieved

4 Awards

- 2016 - 2020, **NUS**, Science & Technology Undergraduate Scholarship
- 2015, **Singapore government**, China-Singapore SM2 Scholarship
  - One of the 90 scholars selected from best High schools in China to study in NUS
- 2013, **Chinese Physical Society**, National High School Physics Competition 3rd prize
- 2013, **Huawei Technologies**, Huawei High School Science Fair best performance award
  - Built Power Electronics based Musical Tesla Coil and Plasma Speaker

5 Skills

- **Physics**: Electrodynamics, Quantum Optics, Quantum Computing, Density Functional Theory
- **Biophysics**: Molecular Dynamics, Gene regulation, Fluid Dynamics, Statistical Mechanics
- **Math**: Linear Algebra, Differential Equations, Probability & Stochastic Processes, Complex Analysis
- **Electronics**: ESP32, Arduino, Raspberry Pi, FPGA, Power & RF Electronics; Altium Designer, Quartus
- **Programming**: Python, JavaScript & Typescript, C & C++, Verilog; Mathematica, MATLAB
- **Frameworks**: PyTorch, TensorFlow, JAX; Node, React, MobX
- **Design**: Solidworks, Autodesk Fusion 360